

Physics Mock Test

Dual Nature of Radiation and Matter

Class 12 | UNIT 7

Marks: 24

Time: 60 Minutes

Coverage: Photoelectric Effect, Work Function, Einstein's Equation, de Broglie Hypothesis, and Wave-Particle Duality.

Constants: $h = 6.63 \times 10^{-34} \text{ Js}$, $c = 3 \times 10^8 \text{ m/s}$, $e = 1.6 \times 10^{-19} \text{ C}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$.

Section 1: Multiple Choice Questions (1 Mark Each)

1. Quantum Nature Evidence

The photoelectric effect provides definitive evidence for which nature of electromagnetic radiation?

- a) Wave nature
- b) Particle nature
- c) Scalar nature
- d) Longitudinal nature

2. K.E. Max Relation

According to Einstein's photoelectric equation, the maximum kinetic energy (K_{max}) of emitted electrons is:

- a) $h\nu - \phi$
- b) $h\nu + \phi$
- c) $\phi - h\nu$
- d) $h\nu / \phi$

3. Threshold Frequency Definition

The threshold frequency (ν_0) is the characteristic minimum frequency of incident radiation below which:

- a) Photoelectrons have zero velocity
- b) No photoemission occurs
- c) Stopping potential becomes positive
- d) Saturation current is maximum

4. de Broglie Relation

The wavelength (λ) associated with a particle of momentum p is given by the de Broglie equation:

a) $\lambda = h / p$

b) $\lambda = p / h$

c) $\lambda = h \cdot p$

d) $\lambda = 1 / (hp)$

5. Experimental Confirmation

Which of the following experiments directly confirms the wave nature of electrons through diffraction?

a) Photoelectric effect

b) Davisson-Germer experiment

c) Stern-Gerlach experiment

d) Millikan's oil drop experiment

6. Stopping Potential Units

If the maximum kinetic energy is 5 eV, the corresponding stopping potential required to halt the photoemission current is:

a) 5 Volts

b) 5 Electron-Volts

c) 0.5 Volts

d) 1/5 Volts

Section 2: Numerical & Short-Answer (2-3 Marks Each)

7. Energy of Photoelectrons

Light of wavelength 400 nm is incident on a metal surface with a work function of 2.5 eV. Calculate the maximum kinetic energy of the photoelectrons ejected (in eV).

8. Determining Threshold Limit

The work function of a certain metal is 4.0 eV. Find the threshold wavelength (λ_0) for this metal. Will visible light (400-700 nm) cause photoemission?

9. Electron Wave Mechanics

An electron is accelerated from rest through a potential difference of 100 V. Calculate its associated de Broglie wavelength.

10. Dual Nature Conceptual

Explain why a moving tennis ball does not show wave-like properties in daily life, whereas an electron does. Calculate the wavelength of a proton ($m_p = 1.67 \times 10^{-27}$ kg) with K.E. = 1 eV.

11. Potentials and Energetics

In a photoelectric experiment, the stopping potential is measured to be 2.0 V. Calculate the maximum speed of the photoelectrons as they leave the metal surface.

12. Work Function Analysis

Incident radiation of frequency 6×10^{14} Hz results in photoelectrons with $K_{\text{max}} = 1.0$ eV. Calculate the work function of the metal in Joules and eV.

Answer Key & Solutions

1. b) Particle nature

Light acts as packets of energy (photons).

2. a) $h\nu - \phi$

Conservation of energy: Energy of photon = Work function + K.E.

3. b) No photoemission occurs

Incident energy $< \phi$.

4. a) $\lambda = h / p$

$p = mv$ for non-relativistic particles.

5. b) Davisson-Germer experiment

Observed electron diffraction in nickel crystals.

6. a) 5 Volts

$eV^s = K_{\max}$. If $K_{\max}$ is X eV, V^s is X Volts.

7. 0.61 eV

$E(\text{photon}) = 1242 / 400 \approx 3.11 \text{ eV}$.

$K_{\max} = 3.11 \text{ eV} - 2.5 \text{ eV} = 0.61 \text{ eV}$.

8. 310.7 nm | No

$\lambda = hc / \phi = (1242 \text{ eV}\cdot\text{nm}) / 4 \text{ eV} = 310.5 \text{ nm}$. Visible light ($\lambda > 400\text{nm}$) has too little energy.

9. 1.227 Å (0.123 nm)

$\lambda = 12.27 / \sqrt{V} \text{ Å} = 12.27 / \sqrt{100} = 1.227 \text{ Å}$.

10. Wavelength $2.87 \times 10^{-11} \text{ m}$

Tennis ball's mass is too large, making λ (h/mv) extremely small and undetectable.
For proton: $\lambda = h / \sqrt{2mK} \approx 2.87 \times 10^{-11} \text{ m}$.

11. $8.39 \times 10^5 \text{ m/s}$

$K_{\max} = 2 \text{ eV} = 3.2 \times 10^{-19} \text{ J}$.

$\frac{1}{2}mv^2 = 3.2 \times 10^{-19} \rightarrow v = \sqrt{(2 \times 3.2 \times 10^{-19} / 9.1 \times 10^{-31})} = 8.39 \times 10^5 \text{ m/s}$.

12. 1.48 eV (2.38×10^{-11} J)

$$E_{\text{photon}} = hf = 6.63 \times 10^{-34} \times 6 \times 10^{14} = 3.978 \times 10^{-19} \text{ J}.$$

$$\phi = E - K = (3.978 \times 10^{-19}) - (1.6 \times 10^{-19}) = 2.378 \times 10^{-19} \text{ J} \approx 1.48 \text{ eV}.$$

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